PapayaPulse: The Smart Farming Framework for Data-driven Papaya Cultivation

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***Abstract*— Papaya cultivation is an important agricultural activity in Sri Lanka, but farmers often face challenges such as inaccurate fruit grading, improper harvest timing, disease management issues, and unfair market pricing. These problems mainly arise due to the lack of accessible technological solutions and reliance on manual decision-making. This research proposes *PapayaPulse*, a mobile-based smart farming framework designed to support data-driven decision-making in papaya cultivation. The system integrates four main components: fruit quality grading using image processing techniques, growth stage detection and harvest prediction using machine learning models, disease and pest identification with severity assessment and rule-based treatment recommendations, and market price prediction for better selling decisions. The application is designed as a user-friendly mobile solution with Sinhala and English language support, making it suitable for local farming communities. By combining multiple AI-based approaches into a single platform, the system aims to assist farmers in improving crop quality, reducing losses, and making informed decisions. The proposed solution highlights the potential of smart farming technologies in enhancing productivity and sustainability in papaya cultivation.**

Keywords— Smart farming; Papaya cultivation; Decision support system; Mobile application; Precision agriculture

I. Introduction

Papaya (*Carica papaya L.*) is one of the most widely cultivated tropical fruit crops in Sri Lanka due to its high nutritional value, rapid growth cycle, and strong demand in both local and export markets [1]. It is primarily cultivated by small and medium-scale farmers in home gardens and small agricultural lands, while commercial cultivation is gradually increasing. Popular hybrid varieties such as Red Lady have demonstrated high yield potential under suitable agronomic and environmental conditions. However, despite these advantages, papaya farmers often face difficulties in maintaining consistent productivity, quality, and profitability due to various practical and environmental challenges.

One of the major challenges in papaya cultivation is the high vulnerable of crops to diseases and pest infestations throughout the growth cycle. Common diseases include fungal infections such as anthracnose and viral diseases such as papaya ringspot virus, while pests such as mites and mealybugs also cause significant damage [2]. When these issues are not detected early, they can spread rapidly and lead to severe yield and quality losses. In most rural farming environments, disease identification and crop monitoring rely heavily on manual observation and farmer experience, which can result in delayed diagnosis and ineffective treatment decisions. Previous studies have shown that deep learning techniques, particularly CNN-based models, can significantly improve disease detection accuracy compared to traditional methods [8], [9], [12].

In addition to cultivation-related challenges, papaya farming also experiences significant post-harvest losses. Improper harvesting time, lack of standardized quality grading systems, and poor handling practices contribute to fruit damage, uneven ripening, and spoilage during transportation and storage. Manual grading based on visual inspection of color, size, and surface condition is often subjective and inconsistent, especially among inexperienced farmers. Research indicates that computer vision techniques can improve grading consistency and reduce human error by analyzing visual features more accurately [3].

Another maor issue faced by farmers is the lack of access to accurate and transparent market price information. Farmers often depend on intermediaries or middlemen to sell their produce without having a clear understanding of actual market value. This results in unfair pricing and reduced income. Studies on agricultural price prediction using machine learning techniques highlight the potential of predictive models to estimate market prices based on multiple factors such as quality, season, and demand [15]. However, these systems are rarely integrated with crop condition data.

With the advancement of technology, smart farming, also known as precision agriculture, has emerged as a promising solution to address these challenges. Technologies such as artificial intelligence, computer vision, and machine learning can automate tasks such as disease detection, crop monitoring, yield prediction, and market analysis [10], [13], [14]. However, many existing solutions focus on isolated tasks and require advanced infrastructure, making them less suitable for smallholder farmers in developing regions [6], [7].

To overcome these limitations, this research proposes PapayaPulse, an integrated mobile-based smart farming framework specifically designed for papaya cultivation in Sri Lanka. The system combines four main components into a single decision support platform: automated fruit quality grading, growth stage recognition and harvest prediction, disease detection with severity-based recommendations, and market price prediction. By integrating these functionalities into a user-friendly mobile application, PapayaPulse aims to provide accurate and explainable insights for both farmers and consumers, improving productivity, reducing losses, and enhancing transparency across the papaya value chain.

II. Literature Review

The application of artificial intelligence (AI) in agriculture has significantly expanded in recent years, driven by the need to improve efficiency, reduce crop losses, and enable data-driven decision-making. Studies have shown that machine learning and deep learning techniques can outperform traditional farming methods in tasks such as crop monitoring, disease detection, and yield prediction [10], [13]. In tropical fruit cultivation, including papaya, AI-based systems have been widely explored to automate processes that traditionally depend on human expertise.

Computer vision techniques, particularly Convolutional Neural Networks (CNNs), have been extensively applied for fruit quality assessment. According to Chuquimarca et al. [3], CNN-based models can effectively analyze visual features such as color, texture, and surface defects to determine fruit quality and maturity. Similarly, deep learning approaches have been widely used for plant disease detection. Early studies by Sladojevic et al. [8] and Mohanty et al. [9] demonstrated the effectiveness of CNN models in identifying plant diseases from leaf images. More recent work by Ferentinos [12] further confirmed that deep learning models can achieve high accuracy across multiple plant disease categories.

Recent advancements have introduced more sophisticated architectures and explainable AI techniques. Gani et al. [4] proposed an attention-guided CNN model for papaya disease detection that incorporates explainability methods such as Grad-CAM and LIME to highlight important image regions influencing predictions. Similarly, de Moraes et al. [5] developed a YOLO-based model for papaya disease and defect detection using a large-scale dataset, demonstrating strong performance in real-world scenarios. Explainable AI has been identified as a critical requirement in agricultural systems, as it improves user trust and understanding of model predictions [6].

Beyond image-based analysis, machine learning models have also been used for predicting crop growth, harvest timing, and market prices. Opara et al. [7] highlighted that models such as Random Forest, neural networks, and regression techniques can effectively handle multiple agricultural variables, including weather conditions and soil properties. Liakos et al. [14] also emphasized the growing role of machine learning in precision agriculture, particularly in integrating environmental and crop data for better decision-making. In the context of price prediction, Goli et al. [15] demonstrated that machine learning models can provide accurate price forecasts based on historical and seasonal data.

Despite these advancements, most existing research focuses on solving a single problem independently. For example, some systems are designed solely for disease detection [8], [9], while others focus on grading or price prediction. This lack of integration forces farmers to rely on multiple tools, reducing usability and limiting adoption, especially among smallholder farmers in developing regions [6], [7]. Additionally, many systems lack practical deployment considerations and are not tailored to real-world farming environments [11].

Another important gap is the lack of support for market transparency and consumer decision-making. While most research focuses on improving production efficiency, very few systems address the needs of consumers in verifying product quality and price fairness. This creates an imbalance in the agricultural value chain and reduces trust between producers and buyers.

In summary, although existing studies demonstrate the effectiveness of AI techniques in improving individual aspects of agriculture, there is still a need for an integrated system. A unified framework that combines multiple functionalities, provides explainable outputs, and supports both farmers and consumers is essential for improving decision-making across the entire papaya farming lifecycle.

III. Methodology

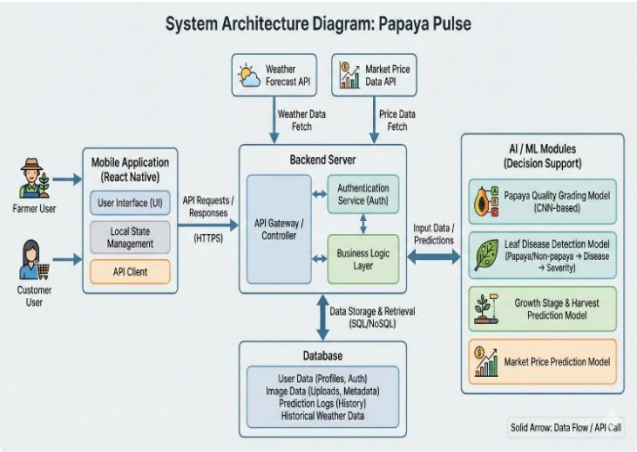

Fig 1: System Architecture Diagram

PapayaPulse is implemented as an integrated, mobile-based smart farming framework that supports papaya cultivation from early crop monitoring to post-harvest market decision-making. The methodology is designed as a unified pipeline where farmers and customers interact with a single application to capture field data, submit observations, and receive intelligent predictions with actionable guidance. The system combines image-based analysis, structured agricultural inputs, environmental context, and trained machine learning models to deliver reliable and explainable outputs. Although PapayaPulse consists of multiple functional components, all components follow a common workflow and are tightly integrated to reflect real-world papaya farming practices.

A. Papaya Quality Prediction and Grading

Papaya quality prediction in PapayaPulse is designed to replace subjective and experience-based grading practices with a consistent, automated, and multi-factor decision mechanism. In traditional papaya trading, quality assessment often varies between farmers, buyers, and middlemen, leading to inconsistent pricing and uncertainty. This component addresses that issue by combining image-based analysis with contextual farming factors to generate transparent and explainable quality grades.

On the farmer side, the system supports two distinct grading workflows: best-quality papayas and factory outlet papayas. For best-quality grading, farmers are required to capture a clear image of the papaya fruit using the mobile camera and provide structured inputs such as district, papaya variety, maturity level (unmatured, half-matured, or matured), number of days since picking. These inputs reflect how experienced farmers traditionally assess quality (1,2,3) and are therefore incorporated into the grading logic.

The captured fruit image is analyzed using trained deep vision models, including Convolutional Neural Networks (CNN) and Vision Transformer (ViT) architectures, to extract visual characteristics such as color distribution, surface uniformity, texture patterns, and visible defects. These image-derived features are then combined with numeric and categorical inputs to generate a final quality grade along with a confidence score. This fusion-based approach ensures that grading decisions are not solely dependent on appearance but also consider freshness, maturity, and handling conditions.

For factory outlet papayas, the grading process focuses on identifying damage severity and processing suitability. Farmers capture images highlighting damaged areas, and the system analyzes defect patterns and surface irregularities to determine whether the fruit should be sold urgently and at what quality level (A, B). Based on the detected condition, urgency-based guidance is provided to reduce post-harvest losses.

On the customer side, the quality prediction workflow is intentionally simplified to support fast and practical decision-making during purchasing. Customers upload a papaya image and input the fruit weight. The system evaluates visual indicators such as color tone and ripeness level and estimates expected taste using temperature-based logic derived from environmental conditions during the picking period. The output includes the predicted quality grade, taste level (e.g., higher taste or lower taste), and a short explanation in clear language. This enables customers to verify fruit quality independently and promotes fairness in local markets.

B. Growth Stage Recognition and Harvest Prediction

Papaya growth detection and first harvest prediction form the central crop-planning component of the PapayaPulse framework. This component is specifically designed to support smallholder and inexperienced farmers by providing clear, data-driven insights into plant development and early harvesting expectations. By integrating visual plant analysis with structured farm inputs and environmental data, the system minimizes uncertainty in both growth assessment and harvest planning.

The component consists of two main functionalities: papaya growth detection and papaya first harvest prediction. In the growth detection workflow, farmers initiate the process by selecting the growth detection option within the mobile application, which automatically activates the device camera. The system provides step-by-step on-screen guidance (in both Sinhala and English) to assist users in correctly capturing a full image of the papaya tree. Once the image is captured, the system first verifies whether the detected object is a papaya plant. If confirmed, a trained image classification model analyzes visual features such as canopy structure, leaf density, stem characteristics, and fruit visibility to determine the plant's current growth stage, categorized into four predefined stages (A, B, C, and D).

Following stage detection, the system delivers human-readable, explainable AI (XAI)-based recommendations to guide farmers toward progressing to the next growth stage. These recommendations include actionable insights such as fertilizer application, soil maintenance practices, watering strategies, and other agronomic factors influencing plant development. By focusing on clear explanations rather than only predictions, the system enhances usability and supports informed decision-making, particularly for farmers with limited agricultural experience.

The second functionality, papaya first harvest prediction, is designed to estimate the timing of the initial harvest rather than the entire production cycle. This approach is specifically tailored to align with agricultural department guidelines commonly followed by smallholder farmers. When users select this option, they are guided to a structured input form where they provide key cultivation details, including the number of papaya trees, watering method and frequency, planting month (approximate), and cultivation district. The system integrates this information with historical environmental data obtained through the Meteo Weather API, including temperature and rainfall patterns corresponding to the selected district and planting period.

A regression-based predictive model processes these combined inputs to estimate the expected timing of the first harvest for all trees in the farm. The output is presented along with explainable, human-readable insights (in Sinhala and English), highlighting how factors such as planting time,

rainfall trends, and temperature conditions influence the predicted outcome. This transparency improves farmer trust and understanding of the system's predictions.

To the best of current knowledge, this component introduces a novel approach by combining automated papaya growth detection with first harvest prediction within a single, user-friendly workflow. By focusing specifically on early-stage decision support and first harvest estimation, PapayaPulse provides practical and accessible guidance for smallholder farmers, enabling proactive crop management and reducing reliance on guesswork throughout the cultivation process.

C. Disease Detection with Severity-Based Recommendations

The disease detection component of PapayaPulse is designed to support early identification of papaya plant diseases that commonly affect cultivation in Sri Lanka and to provide severity-level (mild, moderate, severe) decision support for timely intervention. This functionality aims to reduce crop loss caused by delayed diagnosis and inappropriate treatment by enabling farmers to assess plant health using simple image capture through a mobile device.

Farmers capture images of papaya leaves using the mobile camera, ensuring that visible symptoms such as discoloration, spots, curling, or surface damage are clearly shown. These images are processed using trained deep learning–based image recognition models that analyze visual patterns associated with common papaya diseases. The system is capable of identifying major diseases affecting papaya cultivation, including anthracnose, leaf curl disease, mite infestation, and papaya mosaic. Visual features such as lesion shape, color variation, texture changes, and spread patterns are used to differentiate between disease types.

Beyond disease classification, PapayaPulse performs severity estimation, which enhances the system's ability to evaluate the extent of infection and potential impact on crop yield. By identifying whether the disease condition is mild, moderate, or severe, the system can determine the urgency of intervention. This allows farmers to take preventive measures during early stages while prioritizing immediate treatment for advanced infections, helping reduce unnecessary chemical use and supporting sustainable farming practices.

In addition to analyzing current plant conditions, the system incorporates weather-aware prediction using trained machine learning models. Environmental factors such as temperature, humidity, and rainfall trends are analyzed to estimate the likelihood of disease development or spread in the near future. By integrating weather-based risk assessment, PapayaPulse can alert farmers about potential upcoming disease threats and recommend preventive actions before severe outbreaks occur.

The recommendation component of the system is powered by a hybrid approach combining machine learning models and an agricultural knowledge base. Based on the detected disease, severity level, and environmental conditions, the system generates tailored recommendations that include both organic and inorganic treatment methods. The recommendations also provide practical guidance such as dosage levels, application methods, and treatment intervals to ensure safe and effective use.

All outputs are presented in a structured and easy-to-understand format, including the detected disease name, confidence level, severity classification, future risk indicators, and treatment recommendations. By integrating visual disease detection, severity analysis, weather-based prediction, and intelligent advisory support, PapayaPulse enables farmers to make informed decisions and improve papaya crop management

D. Market Price and Harvest Value Prediction

Market price prediction in PapayaPulse is designed to support both farmers and customers by improving transparency, reducing price uncertainty, and enabling informed economic decision-making across the papaya value chain. This component addresses one of the most critical challenges faced by small and medium-scale farmers lack of access to fair and reliable market price information and simultaneously empowers consumers to verify price fairness before purchase.

On the farmer side, the system assists in planning profitable selling strategies by estimating expected market prices and harvest value. Farmers provide harvest-related inputs such as district, papaya variety, quality grade, total harvest quantity, average fruit weight, and an expected selling day. These inputs reflect the real factors farmers consider when deciding when and where to sell their produce. Using trained machine learning models, the system predicts the fair market price per kilogram for the selected selling day and estimates the total expected income based on the predicted price and harvest volume. In addition, the system identifies the best selling day within a short future window, helping farmers maximize profit by selecting the most favorable time to sell.

For factory-quality papayas, which are highly perishable and often sold at reduced prices, PapayaPulse generates urgency-aware recommendations. When lower quality grades are detected, the system emphasizes rapid selling decisions by providing guidance such as "sell today" or "sell within 1–2 days" to minimize post-harvest losses. This functionality is particularly valuable for new farmers who may lack experience in handling factory outlet produce and time-sensitive selling conditions.

On the customer side, PapayaPulse promotes fairness and trust in local markets by enabling price verification. Customers upload a papaya image and enter the price per kilogram quoted by the seller. The system compares this value against predicted market price ranges derived from the same pricing model used for farmers. Based on this comparison, the price is categorized as fair, overpriced, or underpriced, and a short explanation is displayed to help customers understand the decision. This feature allows buyers to make informed purchasing decisions and discourages unfair pricing practices.

By supporting both producers and consumers within the same framework, the market price prediction component strengthens transparency, reduces dependence on intermediaries, and contributes to a more balanced and trustworthy papaya marketplace.

E. Model Training, Evaluation, and Deployment Strategy

The predictive intelligence of PapayaPulse is implemented using a combination of modern mobile development technologies and machine learning models selected based on task suitability and performance. The mobile application is developed using React Native, enabling cross-platform deployment while maintaining a consistent user experience for both farmers and customers. All prediction logic is designed to be executed through backend services, ensuring that the mobile application remains lightweight and suitable for low-end devices commonly used in rural environments.

For image-based recognition tasks such as papaya quality grading, growth stage identification, and disease detection, deep learning models based on Convolutional Neural Networks (CNN) and Vision Transformer (ViT) architectures were trained using annotated papaya image datasets. These models were selected due to their strong capability in learning visual patterns such as color variation, texture changes, and defect regions without manual feature engineering.

For structured data-driven tasks, different machine learning techniques were applied depending on the prediction objective. Random Forest Classification was used for categorical decision-making tasks such as growth stage classification, quality category selection, disease severity estimation, and price fairness labeling, as it effectively handles mixed numerical and categorical inputs. For numeric predictions including harvest yield, remaining harvest days, market price per kilogram, and income estimation, Random Forest Regression, Linear Regression, and XGBoost Regression models were evaluated. Among these, the models with the most stable and accurate performance on unseen test data were selected for deployment.

All datasets were divided into 80% for training and 20% for testing to ensure unbiased evaluation. During training, multiple candidate models were compared for each task, and the final model was selected based on achieved accuracy, prediction consistency, and alignment with real farming logic rather than purely statistical performance. This ensured that output remains practical and meaningful for end users.

PapayaPulse follows a modular deployment strategy, where each trained model is exposed as an independent backend service through REST-based API endpoints. The mobile application maintains predefined placeholders for these endpoints, allowing trained models to be updated or replaced without modifying the user interface or user workflows. This architecture supports scalability, maintainability, and future system evolution as more data becomes available.

IV. Results And Discussion

The PapayaPulse framework was evaluated as an integrated smart farming system to assess its effectiveness in supporting decision-making across the papaya cultivation lifecycle. The evaluation focused on predictive accuracy, model performance, explainability, and practical usability in real-world farming conditions. Unlike traditional evaluations that analyze individual components separately, this study considers the system as a unified platform.

A. System Performance and Predictive Accuracy

All predictive models were trained using 80% training and 20% testing data split. During development, multiple algorithms were trained and compared for each task, and the best-performing model was selected based on test-set performance and practical stability.

For image-based classification tasks, including papaya quality grading, growth stage recognition, and disease identification, the selected deep learning models achieved an average classification accuracy of approximately 87% on the testing dataset when images were captured according to in-application guidance. Comparative experiments showed that CNN- and Vision Transformer (ViT)–based models consistently outperformed traditional machine learning classifiers for visual recognition tasks, particularly in detecting color variation, texture patterns, and disease symptoms.

For numeric prediction tasks such as harvest timing, yield estimation, and market price forecasting, regression-based models produced stable and realistic outputs. Among the evaluated approaches, Random Forest and XGBoost regression models demonstrated lower prediction error and more consistent trends compared to simple linear regression. Although exact numeric values may vary due to external influences such as unexpected weather changes or sudden market fluctuations, the predicted ranges were found to be sufficiently accurate for practical farm and market decision-making.

These results confirm that model selection based on comparative evaluation, rather than assumption, significantly improves prediction reliability in real-world agricultural scenarios.

B. Explainability, User Trust, and Practical Acceptance

Beyond predictive accuracy, explainability was treated as a core evaluation criterion. Instead of presenting outputs as black-box numerical values, PapayaPulse generates human-readable explanations describing how major contributing factors such as planting month, rainfall patterns, temperature trends, fruit maturity level, or selected selling day influenced each prediction.

To assess practical acceptance, the prototype application was demonstrated to a small group of real papaya farmers and agricultural users during informal feedback sessions. Participants interacted with the application workflows, reviewed prediction results, and evaluated the clarity of explanations and recommendations. Feedback indicated that users were more confident in system outputs when explanations and suggested actions were presented alongside predictions, rather than raw numbers alone.

Farmers particularly valued features that combined visual evidence with contextual reasoning, such as quality grades explained using fruit appearance and freshness

indicators, and harvest predictions justified using seasonal and weather-related information. This observation highlights that explainable outputs play a significant role in building trust and encouraging adoption of AI-based decision-support systems in agriculture

C. Practical Usability and Integrated Value

The first mobile, step-guided interaction design proved effective for users with limited technical experience. The consistent workflow (input → analyze → result → confirmation) reduced navigation errors, while dropdown-based inputs minimized data-entry mistakes. Camera-based evidence collection simplified tasks that are traditionally difficult to assess manually, such as disease detection and quality grading.

Evaluating PapayaPulse demonstrated its ability to support the entire papaya lifecycle within a single platform. Farmers could monitor plant growth, plan harvest timing, assess fruit quality, detect diseases early, and make informed market decisions without relying solely on experience or switching between multiple tools. This integration of production-side intelligence with market-side decision support represents a key strength of the framework.

D. Limitations and Observations

While the evaluation results demonstrate strong potential, several limitations were observed. Prediction quality depends on the accuracy of user-provided inputs and the clarity of captured images. Poor image quality or incorrect field data can reduce model performance. Additionally, market price predictions may be affected by sudden demand changes that are not fully captured by historical datasets.

These limitations highlight opportunities for future improvements through continuous data updates, expanded datasets, and user training. Nevertheless, the evaluation confirms that PapayaPulse provides a reliable, explainable, and practical smart farming solution that offers clear advantages over manual and experience-based practices in papaya cultivation.

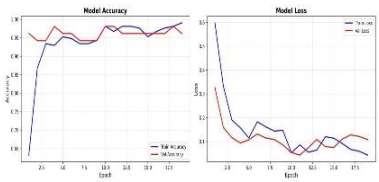

Fig 2: Graphical Representation

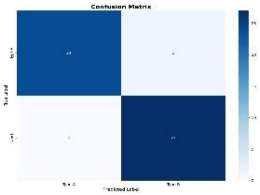

Fig 3: Confusion Matrix

V. Conclusion

This study presented PapayaPulse, a mobile-based smart farming framework designed to improve decision-making in papaya cultivation. The system integrates multiple components, including fruit quality grading, growth stage prediction, disease detection with severity-based recommendations, and market price prediction. These functionalities were developed using machine learning and deep learning techniques such as CNN, Vision Transformer, and regression models. The results demonstrate that the system provides reliable predictions, improves farming efficiency, and supports better market decisions. In the future, the system can be enhanced by expanding datasets, improving model accuracy, and integrating real-time environmental and market data to further support smart agriculture.

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